

Tensor Analysis

10. Christoffel's Symbols

The following symbols

$$[pq, r] = \frac{1}{2} \left(\frac{\partial g_{\alpha\beta}}{\partial p^q} + \frac{\partial g_{\sigma}}{\partial x^p} - \frac{\partial g_{p\beta}}{\partial x^r} \right)$$
$$\left\{ \begin{smallmatrix} s \\ pq \end{smallmatrix} \right\} = g^{nv} [pq, r]$$

are called the Christoffel symbols of the first and second kind, respectively. Other symbols used instead of $\left\{ \begin{smallmatrix} s \\ pq \end{smallmatrix} \right\}$ are (pq, s) and Γ_{pqj} . The latter symbol suggests, however, a tensor character, which is not true in general.

Transformation Laws of Christoffel's Symbols

Suppose we denote by a bar a symbol in a coordinate system x^k . Then

$$\overline{[jk,m]} = [pq,r] \frac{\partial x^p}{\partial \bar{x}^j} \frac{\partial x^q}{\partial \bar{x}^k} \frac{\partial x^r}{\partial \bar{x}^m} + g_{pq} \frac{\partial x^p}{\partial \bar{x}^m} \frac{\partial^2 x^q}{\partial \bar{x}^j \partial \bar{x}^k}$$

$$\overline{\left\{ \begin{matrix} n \\ jk \end{matrix} \right\}} = \left\{ \begin{matrix} s \\ pq \end{matrix} \right\} \frac{\partial \bar{x}^n}{\partial x^s} \frac{\partial x^p}{\partial \bar{x}^j} \frac{\partial x^q}{\partial \bar{x}^k} + \frac{\partial \bar{x}^n}{\partial x^q} \frac{\partial^2 x^q}{\partial \bar{x}^j \partial \bar{x}^k}$$

are the laws of transformation of the Christoffel symbols showing that they are not tensors unless the second terms on the right are zero.